



ESG Constraints and the Markowitz Efficient Frontier

An Empirical Analysis on Russell 3000 Stocks (2017–2022)

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Abstract

This work studies how imposing a constraint on the average portfolio ESG score reshapes the Markowitz efficient frontier. Using a sample of ten Russell 3000 stocks—the five with the highest ESG score and the five with the lowest—over the period January 2017–December 2022, we show that the cost of sustainability is *neither constant nor trivial*. In the sample there is a negative relationship between ESG score and expected return, a necessary condition for the constraint to be *binding* [Azzone et al., 2024]. The constrained frontier coincides with the unconstrained one up to an ESG threshold equal to the natural score of the optimal portfolio (≈ 0.65); above that level, the cost in Sharpe ratio and volatility grows convexly. The composition

analysis shows that the underlying mechanism is a rotation of weights away from Low-ESG stocks (higher expected return) toward High-ESG stocks, with internal heterogeneity in the latter group determining the shape and acceleration of the cost.

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1 Introduction

In recent years sustainable investing has grown at an unprecedented pace. According to Morgan Stanley, in the first half of 2025 sustainable funds reached \$3.92 trillion in assets, outperforming traditional funds with a median return of 12.5% versus 9.2% [Morgan Stanley, 2025]. The growing attention to *Environmental, Social and Governance* (ESG) factors has stimulated a broad literature investigating the impact of such considerations on portfolio choices and on the costs borne by investors [ESMA, 2022].

From a theoretical standpoint, introducing an ESG constraint into the classical mean-variance optimization problem of Markowitz [1952] shrinks the feasible set by construction, pushing the efficient frontier outward. This result is formally obvious: any additional constraint cannot improve (and can only worsen or leave unchanged) the optimum. The economically relevant question is therefore not *whether* the frontier shifts, but:

1. **By how much** it shifts—the “price of sustainability” measured in extra volatility and lost Sharpe ratio;
2. **When** the constraint is *binding*—which depends on the sign of the relationship between ESG score and expected return in the sample;
3. **Why** the frontier shifts—that is, how the composition of the optimal portfolio changes as the ESG threshold varies.

Azzone et al. [2024] show that, in a sample of Russell 3000 stocks over 2017–2022, the ESG constraint is binding only when the relationship between ESG and return is negative, and that the cost grows nonlinearly with the tightness of the constraint. This work aims to replicate and extend that analysis on a restricted subsample of ten stocks, prioritizing the readability of the results and the economic interpretation of the composition.

The remainder of the paper is organized as follows. Section 2 introduces the ESG constraint. Section 3 describes the data and methodology. Section 4 discusses the empirical results. Section 5 concludes.

2 The ESG Constraint

We work in the standard mean-variance setting: N assets with expected-return vector $\boldsymbol{\mu} \in \mathbb{R}^N$ and positive-definite covariance matrix $\boldsymbol{\Sigma}$, and a portfolio described by a weight vector $\boldsymbol{w}$ with $\mathbf{1}^\top \boldsymbol{w} = 1$. For each target return μ_p the efficient portfolio is the one of minimum variance, and the collection of such portfolios traces the efficient frontier.

A sustainability mandate adds a constraint on the weighted-average ESG score of the portfolio. Let $\boldsymbol{s} \in \mathbb{R}^N$ be the vector of asset ESG scores and $\tau \in [0, 1]$ a threshold. The

constrained problem is

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \boldsymbol{\mu}^\top \mathbf{w} = \mu_p, \quad \mathbf{s}^\top \mathbf{w} \geq \tau, \quad (1)$$

to which we add the non-negativity constraint $\mathbf{w} \geq \mathbf{0}$ when short selling is not allowed.

Adding the inequality $\mathbf{s}^\top \mathbf{w} \geq \tau$ shrinks the feasible set, so the constrained frontier can only coincide with or lie outside the unconstrained one. The economically interesting question is whether it actually moves, i.e. whether the constraint is *binding*. If the unconstrained optimum already satisfies $\mathbf{s}^\top \mathbf{w}^* \geq \tau$, the constraint is slack and its cost is zero. The constraint bites only when two conditions hold together: the threshold is high enough to exceed the natural ESG score of the optimal portfolio, and ESG scores and expected returns are negatively related in the data, so that raising the average score forces the investor to give up return. These are precisely the empirical conditions we examine in Section 4.

3 Data and Methodology

3.1 Sources and Coverage

The data come from two datasets:

- **ESG scores:** monthly scores for 3941 ISINs, period January 2017–December 2023, scale 0–100 (normalized to [0,1]). Source: an ESG data provider (scale consistent with that used in Azzone et al., 2024).
- **Returns:** monthly returns for 2727 stocks, period January 2017–December 2022.

The two datasets are matched by ISIN. The common sample comprises 2680 stocks with full coverage. Only stocks with at least 60 monthly return observations are deemed valid.

3.2 Subsample Selection

To keep the analysis interpretable, we select **10 stocks:** the 5 with the highest average ESG score and the 5 with the lowest in the valid universe. This “extreme” selection maximizes the contrast between the two groups and makes the ESG-return relationship evident. Table 1 reports the characteristics of the selected stocks.

Table 1: Characteristics of the ten selected assets (annualized returns, period 2017–2022, $r_f = 3\%$).

ISIN	Ticker	Group	ESG	μ (%)	σ (%)
US6907421019	OC	High-ESG	0.924	13.3	34.2
US5949181045	MSFT	High-ESG	0.912	23.7	20.6
US1941621039	CL	High-ESG	0.903	6.3	16.8
US88579Y1010	MMM	High-ESG	0.894	−4.3	23.1
US0138721065	AA	High-ESG	0.881	29.9	56.1
US4333231029	HIFS	Low-ESG	0.074	8.7	26.2
US6350171061	FIZZ	Low-ESG	0.068	19.0	45.1
US74158E1047	PNRG	Low-ESG	0.066	16.4	46.9
US9706461053	WLFC	Low-ESG	0.065	24.8	39.7
US14817C1071	PTIE	Low-ESG	0.053	31.9	68.7

3.3 The Selected Companies

Identifying the companies behind each ISIN is useful for interpreting the results, because the cost of the constraint ultimately depends on the financial profile of the individual stocks. The High-ESG group is dominated by large, established industrials and consumer companies, while the Low-ESG group collects smaller, more specialized and often more volatile names.

High-ESG group.

- **OC – Owens Corning.** Building-materials industrial that manufactures insulation, roofing and fiberglass composites. A moderately cyclical company tied to the construction cycle.
- **MSFT – Microsoft.** Global software and cloud-computing leader (Productivity, Intelligent Cloud, Personal Computing). High expected return and low volatility make it the best Sharpe ratio in the sample—a key driver of the results.
- **CL – Colgate-Palmolive.** Consumer-staples multinational and leader in oral and household care. A defensive, low-volatility stock whose stability makes it attractive for variance minimization.
- **MMM – 3M.** Diversified industrial conglomerate. Over the sample period it was weighed down by litigation and operating headwinds, resulting in a negative average return despite a high ESG score.
- **AA – Alcoa.** Aluminum producer, a highly cyclical materials and commodities company. High expected return but very high volatility.

Low-ESG group.

- **HIFS – Hingham Institution for Savings.** Small Massachusetts savings bank founded in 1834. A niche community lender with a moderate risk-return profile.
- **FIZZ – National Beverage.** Producer of sparkling waters and soft drinks (LaCroix brand). A consumer company with a relatively volatile return profile.
- **PNRG – PrimeEnergy Resources.** Small-cap oil and gas exploration and production company. An energy stock with high volatility and commodity-price sensitivity.
- **WLFC – Willis Lease Finance.** Specialty-finance company that leases commercial aircraft engines. High expected return with substantial volatility.
- **PTIE – Pain Therapeutics (now Cassava Sciences).** Clinical-stage biotechnology firm developing an Alzheimer’s-disease therapy; it traded as PTIE over the sample before its 2019 rename to Cassava Sciences (ticker SAVA). The highest expected return *and* the highest volatility in the sample—a speculative, binary-outcome name.

3.4 Portfolio Parameters

Parameters are computed on monthly returns and annualized: expected return $\mu_i = 12 \cdot \bar{r}_i^{monthly}$, covariance matrix $\Sigma = 12 \cdot \hat{\Sigma}^{monthly}$. The risk-free rate is set to $r_f = 3\%$, consistent with US Treasury rates over the reference period.

4 Empirical Results

4.1 The ESG–Expected-Return Relationship

Figure 1b plots the relationship between ESG score and annualized expected return for the ten selected stocks, together with the OLS regression line. The relationship is **negative**: the estimated slope is $\hat{b} = -0.078$, with a correlation $\rho = -0.305$.

Although the relationship is not statistically significant (a sample of only 10 observations), the direction is consistent with the results of Azzone et al. [2024] on the full Russell 3000. As discussed in Section 2, a negative association is what makes the ESG constraint *binding* for sufficiently high thresholds and gives it a positive cost.

A relevant aspect is the internal heterogeneity within the two groups. In the High-ESG group, MSFT exhibits a high ESG score (0.912) *and* an attractive expected return

(23.7%), with low volatility (20.6%) and a Sharpe ratio of 1.00—the best in the sample. This feature, shown in Figure 1a, has crucial implications for the composition of the optimal portfolio, as we will show in Section 4.5.

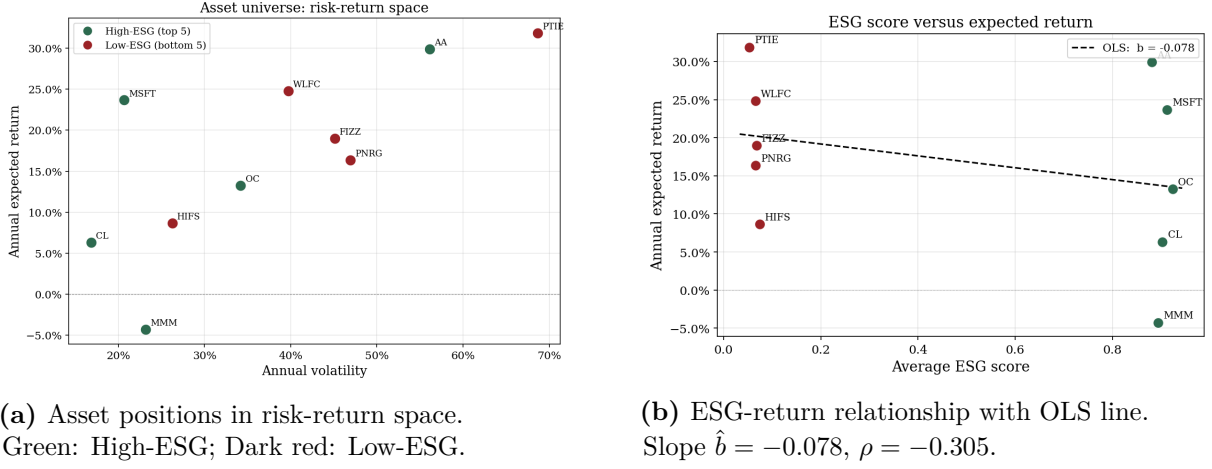


Figure 1: Characteristics of the ten assets and the ESG-return relationship.

4.2 The Reference Efficient Frontier

Before introducing the ESG constraint, we compute the analytic closed-form frontier, which allows short selling. The global minimum-variance portfolio (MVP) has:

$$\mu_{MVP} = 11.3\%, \quad \sigma_{MVP} = 13.4\%, \quad \text{ESG}_{MVP} = 0.77.$$

The ESG score of the analytic MVP is surprisingly high (0.77): the variance-minimization problem, with short selling, naturally loads on low-variance stocks such as CL ($\sigma = 16.8\%$), which also has a very high ESG score (0.903). As a result, the ESG constraint is *non-binding* for thresholds below 0.77 even in the short-selling case.

For the long-only case, the maximum-Sharpe optimal portfolio has:

$$\mu^* = 24.5\%, \quad \sigma^* = 18.3\%, \quad \text{Sharpe}^* = 1.18, \quad \text{ESG}^* = 0.653.$$

The natural ESG score of the long-only optimal portfolio (≈ 0.65) is the effective threshold below which the constraint does not bite.

4.3 Constrained Frontiers: Short Selling vs. Long-Only

Figure 2 shows the most important result of the work in visual form: the effect of the ESG constraint depends *structurally* on the ability to sell short.

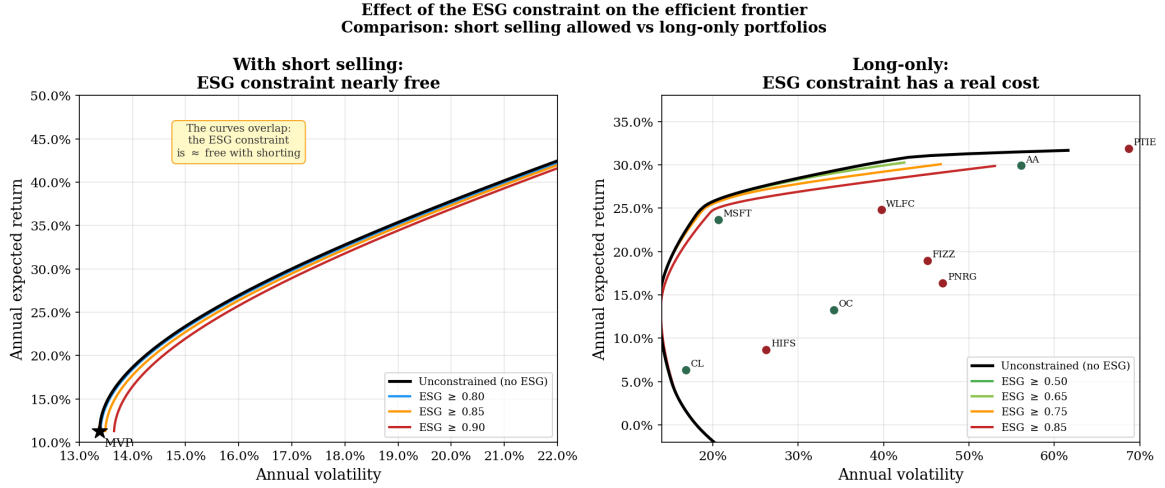


Figure 2: Effect of the ESG constraint on the efficient frontier. *Left:* with short selling the constrained frontiers coincide almost perfectly with the unconstrained one, even for very high thresholds (ESG ≥ 0.90): the constraint is nearly free. *Right:* in the long-only case the curves separate clearly as the threshold rises, with a progressive shift to the right (more volatility for a given return).

With short selling: near-zero cost. With short selling allowed, the ESG constraint is nearly free even for very high thresholds (ESG ≥ 0.90). The reason is intuitive: with short positions one can satisfy the average ESG constraint simply by going short on Low-ESG stocks and long on High-ESG ones, without significantly altering the variance structure of the portfolio. The three constrained MVPs differ from the unconstrained one by only 0.3–0.5 percentage points of volatility. This result confirms the theory: Azzzone et al. [2024] show that the constrained frontier (a tracking-error constrained frontier in the sense of Roll, 1992) collapses onto the unconstrained one when the correlation between ESG and return is not sufficiently negative, or when the additional instrument (short selling) offers enough degrees of freedom.

Long-only: measurable and growing cost. Without the ability to sell short, the ESG constraint creates a clear separation between the frontiers. Figure 3 shows the detail with seven thresholds ($\tau \in \{0.50, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85\}$): each increase in the threshold shifts the frontier to the right, with the most pronounced separation in the upper part of the chart (high returns) where concentration on MSFT is not enough and one is forced to include inefficient stocks such as CL and MMM.

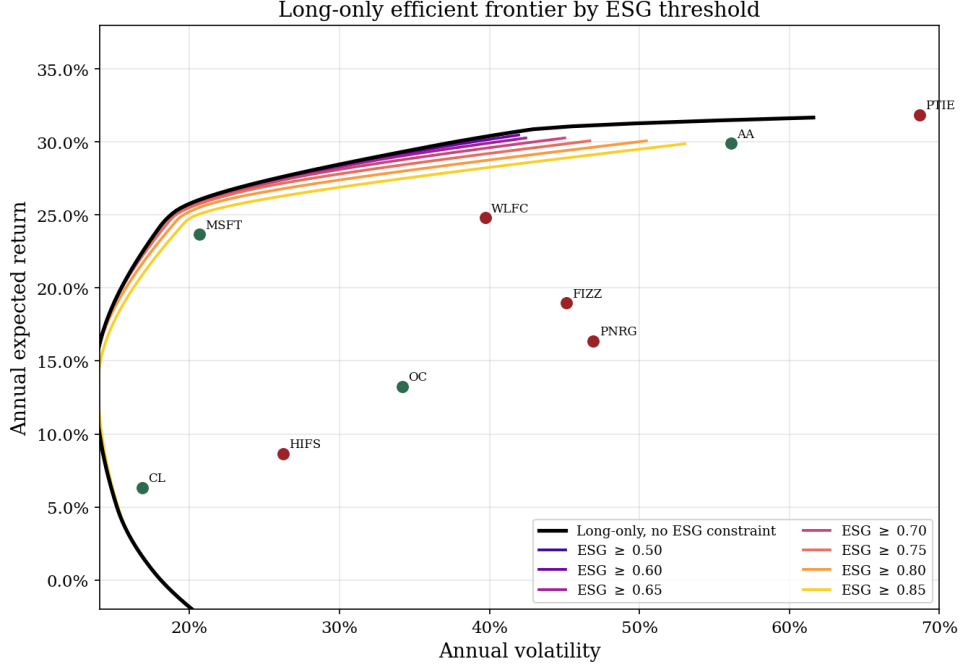


Figure 3: Long-only efficient frontier for seven ESG thresholds. The main kink, around $\mu \approx 25\text{--}26\%$, corresponds to the corner portfolio where MSFT exhausts its capacity to raise the target return without adding riskier assets. The constrained frontiers (high thresholds) end earlier on the right: PTIE, the asset with the highest return ($\mu = 31.9\%$), is excluded because it is Low-ESG.

A visible feature of the long-only case is that the frontier is no longer a smooth curve but bends at a few corners. This is structural rather than a numerical artifact: as the target return rises, the optimal portfolio changes the set of assets it actually holds, and each time an asset enters or leaves, the frontier switches to a different, steeper segment. The most evident corner sits around $\mu \approx 25\%$. Below it the portfolio is dominated by MSFT ($\mu = 23.7\%$, $\sigma = 20.6\%$), the single most efficient stock; once the target exceeds MSFT’s own return, the only way to push returns higher is to add far more volatile names (WLFC, AA, PTIE), so the frontier bends sharply to the right.

The ESG-constrained frontiers (high thresholds) bend earlier and stop sooner on the right. The truncation reflects the exclusion of the highest-return Low-ESG names: with PTIE ($\mu = 31.9\%$) and WLFC ruled out, the maximum return attainable under $\text{ESG} \geq 0.85$ falls to about 30%. This is also why the marginal cost of the constraint is so high at the top of the frontier: it is not only extra volatility, but expected returns that become structurally unreachable.

4.4 The Cost of Sustainability

Figure 4 quantifies the cost of the ESG constraint in two complementary ways.

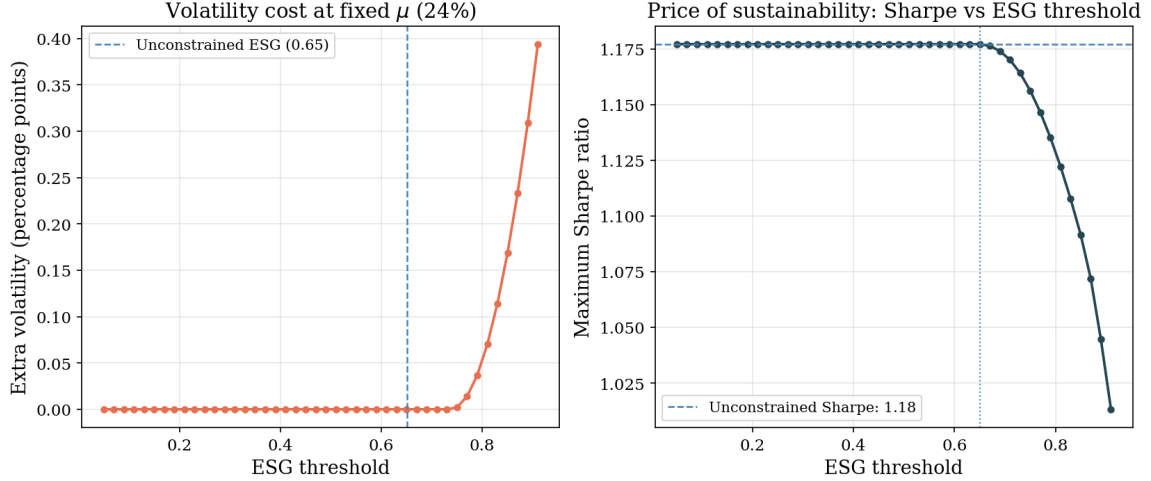


Figure 4: Cost of sustainability as a function of the ESG threshold. *Left:* extra volatility (in percentage points) at a fixed target return of 24.5%. *Right:* maximum Sharpe ratio of the constrained long-only portfolio; the dashed line marks the unconstrained Sharpe (1.18).

The left panel shows the extra volatility needed to maintain a 24.5% return as the ESG threshold rises. The cost is **essentially zero** up to $\tau \approx 0.65$, then grows convexly: each 0.05 increase in the threshold above 0.75 costs about 0.5–1 percentage point of additional volatility.

The right panel reports the maximum Sharpe ratio attainable under the ESG constraint (long-only portfolio). The Sharpe ratio stays stable at 1.18 up to $\tau \approx 0.65$, then declines progressively: with a threshold of 0.90, the Sharpe ratio falls to 1.03, a loss of about 13 basis points.

This convex cost pattern—also illustrated through the shadow price in Figure 5—is the central result of the analysis. As Azzone et al. [2024] argue, it reflects the geometric structure of the frontier: *the first degrees of sustainability come at low cost, the last ones cost a great deal in terms of efficiency.*

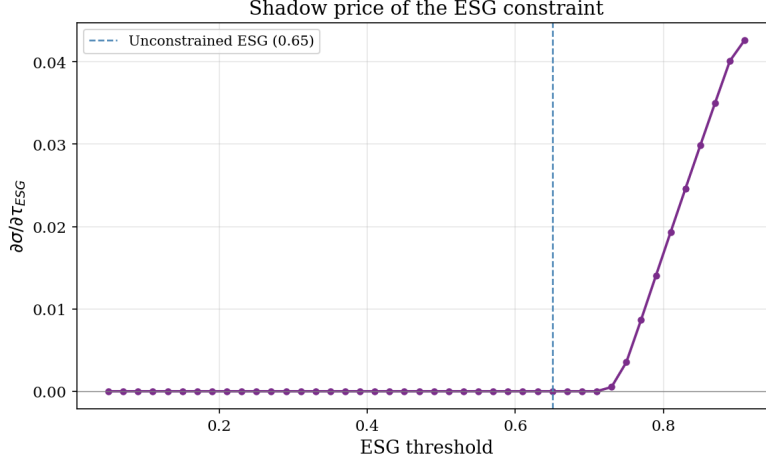


Figure 5: Shadow price of the ESG constraint: numerical derivative of the minimum volatility with respect to the threshold τ , at a fixed return of 24.5%. The shadow price is flat (near zero) for low thresholds and grows rapidly beyond the critical point $\tau^* \approx 0.65$.

4.5 Composition of the Optimal Portfolio

The composition analysis answers the question: *why* does the frontier shift as observed? Figure 6 shows the composition of the maximum-Sharpe portfolio (long-only) for increasing ESG thresholds.

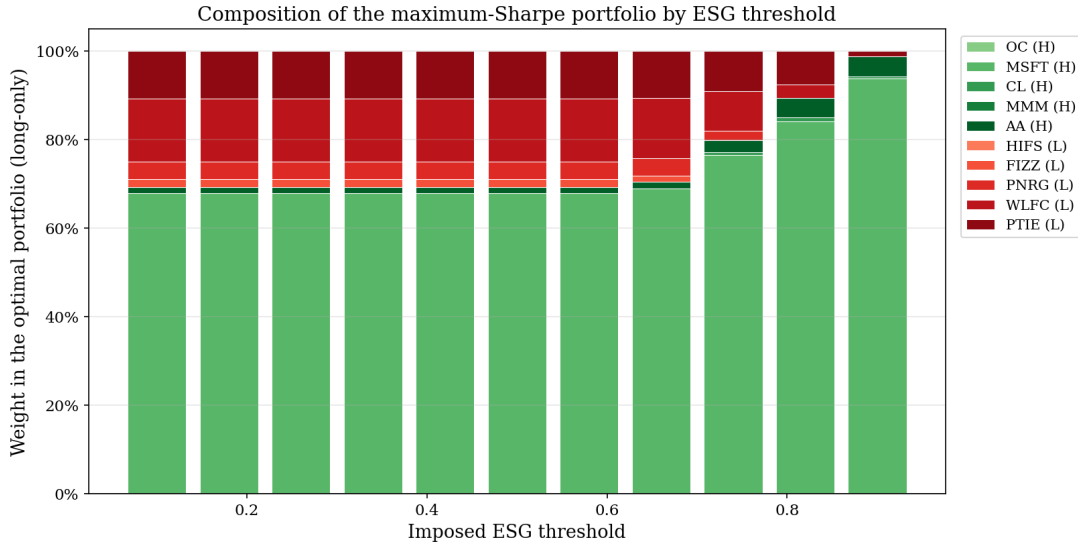


Figure 6: Composition of the maximum-Sharpe portfolio (long-only) as the ESG threshold varies. Green: High-ESG stocks; red: Low-ESG. Darker shades indicate stocks with higher ESG (green group) or lower ESG (red group).

Three relevant features emerge:

1. MSFT is the dominant driver. Microsoft (ticker MSFT) has ESG = 0.912 and an individual Sharpe ratio of 1.00, the highest in the sample. Even without an ESG

constraint the optimal portfolio allocates about 68% to MSFT. As the threshold rises, MSFT’s weight increases further (up to 93% at a threshold of 0.90), since it is the only High-ESG stock with a good risk-return profile.

2. The rotation goes from the Low-ESG group toward High-ESG. The aggregate weight of the Low-ESG group (HIFS, FIZZ, PNRG, WLFC, PTIE) falls from about 30% with no constraint to 1% at a threshold of 0.90. It is this rotation that mechanically explains the shift of the frontier: stocks with high expected return but low ESG (PTIE, WLFC) are sold to buy stocks with high ESG but lower return.

3. The other High-ESG stocks enter only “by necessity”. CL, MMM and OC—although belonging to the High-ESG group—have low expected return ($\mu_{CL} = 6.3\%$, $\mu_{MMM} = -4.3\%$) and are included in the portfolio only when the threshold makes them necessary. This explains the convexity of the cost: for moderate thresholds it is enough to overweight MSFT (nearly free in terms of Sharpe), while for very high thresholds one must acquire CL and especially MMM (negative return), which drastically compresses efficiency.

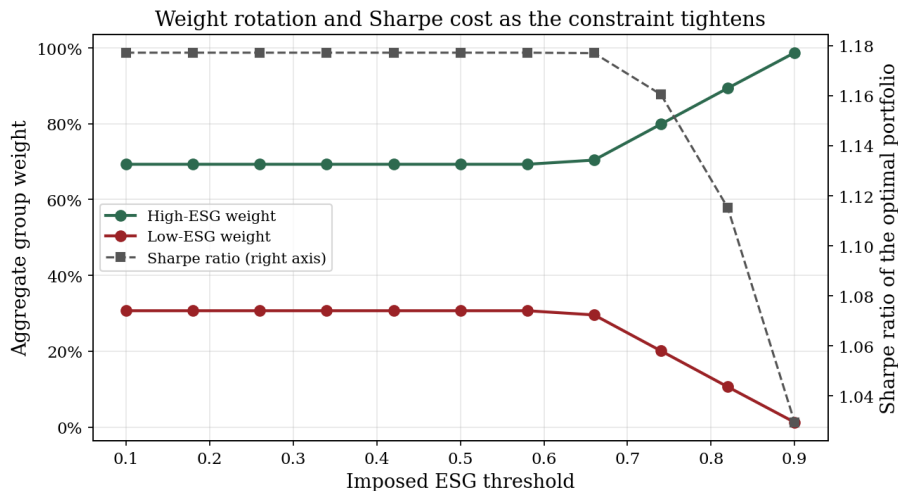


Figure 7: Aggregate High-ESG and Low-ESG weights (left axis) and Sharpe ratio (right axis) as the ESG threshold varies. The critical threshold $\tau^* \approx 0.65$ separates the zero-cost phase from the growing-cost phase.

This internal heterogeneity within the High-ESG group is the richest result of the analysis. The cost of sustainability does not depend only on *how much* ESG is required, but on *which* ESG assets one is forced to include. A universe with more MSFT-type stocks (high ESG *and* high return) would make the constraint nearly free; a universe with only MMM-type stocks (high ESG but negative return) would make it very costly.

5 Conclusions

This work has studied how an ESG constraint reshapes the Markowitz efficient frontier, using a sample of ten Russell 3000 stocks selected at the extremes of the ESG-score distribution (five stocks with the highest and five with the lowest average score over 2017–2022).

The main results are threefold. **First**, in the sample there is a negative relationship (albeit not statistically significant with 10 observations) between ESG score and expected return, a necessary condition for the constraint to be binding. **Second**, the cost of sustainability is zero for ESG thresholds below the natural score of the optimal portfolio (≈ 0.65) and grows convexly above that level: the first degrees of sustainability come at near-zero cost (in Sharpe ratio and volatility), the last ones cost a great deal. **Third**, the underlying mechanism is a rotation of composition away from Low-ESG stocks (high expected return) toward High-ESG stocks, but with strong internal heterogeneity in the latter group: MSFT, with high ESG *and* a high individual Sharpe, is the stock that allows sustainability to have a contained cost at moderate thresholds, while stocks such as CL and MMM (high ESG, low or negative return) enter the portfolio only when the threshold is very tight, driving the sharp increase in cost.

From the standpoint of practical implications, these results suggest that it is not enough to classify stocks as “sustainable” or “unsustainable”: the quality of the risk-return profile within the sustainable group is decisive for the cost the investor pays. From an asset-management perspective, the “free lunch” of sustainability—if it exists—depends on the availability of assets with high ESG *and* high financial efficiency. The composition analysis proposed in this work provides a tool to identify them.

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